

[DC] Towards Comprehensible and Expressive Teleportation Techniques in Immersive Virtual Environments

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ABSTRACT

Teleportation, a popular navigation technique in virtual environments, is favored for its efficiency and reduction of cybersickness but presents challenges such as reduced spatial awareness and limited navigational freedom compared to continuous techniques. I would like to focus on three aspects that advance our understanding of teleportation in both the spatial and the temporal domain. 1) An assessment of different parametrizations of common mathematical models used to specify the target location of the teleportation and the influence on teleportation distance and accuracy. 2) Extending teleportation capabilities to improve navigational freedom, comprehensibility, and accuracy. 3) Adapt teleportation to the time domain, mediating temporal disorientation. The results will enhance the expressivity of existing teleportation interfaces and provide validated alternatives to their steering-based counterparts.

Index Terms: Virtual Reality, Navigation, Teleportation

1 BACKGROUND AND MOTIVATION

Navigation is one of the most fundamental tasks in every VR application. A particularly widespread approach is teleportation, where users egocentrically select a target location before being automatically relocated [4]. Due to the relocation being instantaneous, teleportation is more efficient, especially when traversing large distances. Furthermore, various studies have shown that teleportation-based travel techniques reduce cybersickness as these techniques avoid visual cues of self-motion that can lead to discomfort and are therefore often preferred over continuous techniques [4, 10]. While teleportation is easy to implement and use, it also comes with trade-offs. A major drawback due to the instantaneous relocation is the reduced spatial awareness [4] and navigational freedom, as targets can only be selected where scene geometry is present.

In my thesis, I want to mitigate common drawbacks of teleportation with the goal to make teleportation a more sophisticated technique so it can be used as a replacement for other continuous techniques while ensuring a high *Comprehensibility* of the process, which means that the chosen mechanisms should “foster the awareness of ongoing navigation activities and facilitate the predictability of their consequences” [9]. My thesis consists of three blocks and research questions concerning the *target specification* and *pre-travel information* steps of teleportation [10]. First, I want to assess how different common mathematical models for target selection affect teleportation distance and accuracy. Next, I want to extend the capabilities of teleports to allow for more navigational freedom by 1) improving the comprehensibility of existing techniques for teleporting to mid-air targets and scaling techniques and 2) combining scale and height specification into a single technique. Finally, I want to adapt teleportation to the time domain with suitable pre-travel information to reduce temporal disorientation. I therefore formulated the following research questions:

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- Q1: What are good parametrizations for common mathematical models used to specify the target location?
- Q2: How can we extend teleportation to allow more navigational freedom without amplifying the risk of spatial disorientation?
- Q3: How can teleportation be extended to the time domain in a comprehensive manner?

2 PARAMETRIZATION OF TARGET SPECIFICATION (Q1)

Default teleportation techniques provided by game engines like Unreal or Unity simulate a projectile following the projectile motion equation, which is influenced by the controller height, gravity, and an initial projectile velocity. That projectile’s intersection point with the scene’s geometry then determines the target location. While we do not want to alter the gravity and the controller height is given by the tracking system, it is not clear what values are used as the initial velocity, which directly influences the maximal teleportation distance.

To get more insights into the influence of the initial projectile velocity on the selected distance and accuracy of teleportation, we conducted a within-subjects design user study with 60 participants. In the study, participants were positioned on a river with multiple circular platforms placed in front of them and were then asked to teleport to the farthest platform possible. The farther they teleported, the greater the score they received. Missing a platform, however, resulted in a major point reduction. Participants, therefore, needed to find a balance between far and accurate teleports. In total, we had 10 different conditions with varying velocities (from 6 m/s to 22 m/s in increments of two and an infinite velocity condition to simulate a straight ray). We hypothesized that, although higher velocities would allow farther teleports, there is a projectile velocity threshold beyond which the selected distances will not get larger anymore as hand jitter leads to accuracy problems, and the target platforms become harder to see.

First results indicate that this hypothesis can be confirmed. Interestingly, the teleportation distances decrease for large velocities and the straight ray, indicating that there is an optimal velocity beyond which selection becomes too difficult, resulting in participants selecting closer targets. We also calculated index of difficulty values and compared them to the teleportation distance to get an efficiency measure for teleportation (similar to the throughput value in a traditional Fitts’ Law analysis), which could potentially be used to compare different teleportation techniques. The paper is currently in the writing process and not fully evaluated, but the results will serve as a foundation for future developments of teleportation techniques.

3 IMPROVING NAVIGATIONAL FREEDOM (Q2)

With the first user study establishing good parametrization values for the teleport parabola, we want to investigate advanced teleportation techniques that improve navigational freedom. While prior work exists allowing users to teleport to mid-air targets [7] or to adjust the user’s scale [8], additional challenges regarding the comprehensibility of these more complex techniques arise. In both cases, the user’s initial location differs from the target location not only on a horizontal level but also on a vertical or scale level. Traditionally, horizontal changes are mediated by adding suitable pre-travel

information like e.g., rendering a duplicate of the avatar at the target location to reduce spatial disorientation. However, it is unclear how they can be adapted to mediate vertical and scale changes. In a potential user study, we want to compare different types of pre-travel information like rendering only a sphere, a preview avatar, or a complete preview of the surroundings in the form of a World-in-Miniature or interactive portal further to improve the comprehensibility of teleportation to mid-air targets.

In a second step, we also want to combine height and scale teleportation into a single technique, incorporating the learnings from the previous study. We already developed a simple combination of scale and height, where we implemented the two-step technique from Weissker et al. [7] and added a simple scale technique, where users could press two buttons on the controller to increase/decrease their scale at fixed intervals. The technique was used in a digital twin scenario and published as a workshop paper [6]. While we did a short evaluation that was more about the system in general, the results indicated that this combined technique was well received, which motivates further research into this area.

4 TEMPORAL NAVIGATION (Q3)

So far, teleportation is used mainly for spatial navigation, while navigation through time is usually done via time sliders. However, the issue with these continuous approaches is that these interfaces often suffer from accuracy when locating specific events in a recording [3]. A common solution has been to highlight objects of interest and to display their motion trajectories, allowing users to drag objects along them [3]. While this approach is suitable for visualizing location changes, moving through time along them can create an overwhelming visual flow as all other time-variant objects also shift. Additionally, discrete events that often occur, e.g., in authoring environments, may easily be scrubbed over. This is especially true when multiple changes occur in quick succession. Another method involves breaking continuous recordings down into key moments [2]. However, traveling to these key moments is currently done instantaneously without mediation, resulting in numerous simultaneous changes. Forming an understanding of these changes in the time domain is similar to the formation of spatial awareness in spatial navigation. We therefore developed a novel teleportation-based time navigation technique called *TENETvr*, that assists users in the formation of *temporal awareness*, which we defined analogously to the definition of spatial awareness from Bowman et al. as: “the ability of the user to retain an awareness of her surroundings during and after travel to a *different time point*” [1].

The teleportation with our technique is executed in three steps. 1) Users select an object in the environment via raycast and are then presented with a list of key moments, which are small representations of the selected object at different points in time. These key moments are arranged chronologically in a circular pattern around the user for easy access and to provide a visual overview of the object’s evolution. 2) To specify the desired target time to teleport to, users simply grab one of these key moments. A crucial aspect of *TENETvr* is providing pre-travel information before the time jump. While the user holds the selected key moment, the system visually highlights all changes that occurred in the environment between the current time and the target time. This pre-travel information is designed to reduce temporal disorientation and improve the user’s understanding of the temporal transition. 3) Once the user is ready, they activate the time jump by pressing the trigger button while holding the key moment. The virtual environment instantaneously transitions to the selected time point, reflecting all the changes highlighted during the pre-travel information phase.

In a user study with 20 participants, we evaluated the effectiveness of the added pre-travel information and results showed that with pre-travel information present, participants had a better comprehension of the changes in the environment after the teleportation.

More information is given in our paper [5]. For future research, it might be interesting to look at the mediation of temporal distance since users might not only be interested in a single key moment but also understand the temporal relation to other key moments or how these specific events were triggered. Therefore, The *TENETvr* interface could be adapted to convey temporal distance and allow key moments to be dragged to allow for small continuous adjustments.

5 CONCLUSION

The proposed topics for my thesis advance teleportation techniques for both spatial and temporal navigation. By optimizing the target specification process, extending navigational freedom through combined height and scale adjustments, and extending teleportation to the time domain, key limitations of traditional teleportation methods are addressed to improve comprehensibility. Since I already finished work on Q1 and Q3, I would mostly like to discuss the planned work for Q2, whether it fits into the overall story and what aspects I presented might be worth working on.

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